

K561 — Saturday convergent close: Lyra F358 mass-mixing UNIFICATION via Korányi-Wolf tiers (clean walk-back from “11/20 unclear” + “mass and mixing different structures”; substantive substrate-architectural finding emerges from the walkback) + Lyra HONEST 79-vs-80 flag (Cal-#411-class: leading structure $4/80 = 1/20$; $-1 \rightarrow 79$ is T914-style prime-residue — forced or fit-suspect?; doesn’t bank the prettier 79) + Grace + Elie + Lyra ALL CONVERGE on rank-2 K-type address as unifying open core + Elie “ $4 \rightarrow 9$ potential” framing EXCEEDS Cal’s K560 explicit verdict (“please don’t pre-count it”; Cabibbo is structural-tier LEAD not candidate count-mover) + Keeper holds count at 5/26 per Cal verdict + routing recommendation = three-way paired K-type address build START NOW (not Saturday-closeable but Saturday-startable cleanly) with Cal cold-reads continuing in parallel + no EOD until Casey calls

Date: 2026-06-27 Saturday late afternoon · Auditor: Keeper · Inputs: Lyra F358 walk-back + mass-mixing unification + honest 79-vs-80 flag + Grace substantive close + corpus rediscovery + off-axis + Elie consolidated status + “ $4 \rightarrow 9$ potential” framing + three-way convergence on K-type address + three CIs request routing

Verdict — SUBSTANTIAL Saturday convergent close: Lyra F358 substantive substrate-architectural UNIFICATION (mass and mixing on same three Korányi-Wolf tiers; mass- ν and mixing-N as two coordinates on each stratum; monotonically linked); Lyra HONEST 79-vs-80 Cal-#411-class flag; three-way convergence on rank-2 K-type address as unifying open core; Elie “ $4 \rightarrow 9$ ” framing EXCEEDS Cal’s K560 verdict — Keeper holds count at 5/26

Count: 5 of 26 — held per Cal’s K560 explicit verdict (“please don’t pre-count” Cabibbo); Elie’s “ $4 \rightarrow 9$ potential” trajectory framing is over-projection at count level.

Landing 1 — Lyra F358 mass-mixing UNIFICATION via Korányi-Wolf tiers

The substantive substrate-architectural unification

“Walking back the error produced a real result: with today’s ν -assignments (F354/F355), the mass and mixing sectors are the same three Korányi-Wolf tiers — each generation is one stratum carrying two coordinates: a mass- ν and a mixing-N, monotonically linked (e: $\nu=5/2/N=1$ /origin; μ : $\nu=3/2/N=0.55$ /Cartan; τ : $\nu=0/N=0$ /Shilov). The electron-at-origin is the $\nu=5/2$ BF point. So mass (F345–F355) and mixing (F87b) are one geometry seen two ways.”

Substantive verification (Keeper): - Three Korányi-Wolf strata = three generations (F86 + K548 lineage) - Each generation = one stratum with two coordinates (mass- ν + mixing-N) - Monotonic mapping: e ($\nu=5/2$, $N=1$, origin) $\rightarrow \mu$ ($\nu=3/2$, $N=0.55$, Cartan slice) $\rightarrow \tau$ ($\nu=0$, $N=0$, Shilov boundary) - Electron at BF point $\nu=5/2$ (per F355 positivity theorem) - Mass + mixing = ONE geometry seen via two coordinates per stratum

Per [[feedback_corpus_reconnection_before_declaring_irreducible]] STANDING: Lyra walked back “mass and mixing are different structures” (her K560 honest negative) when corpus + Grace’s discovery showed unification. Substantive substrate-architectural finding emerges FROM the walk-back.

Per Cal #35 STANDING (Schur audit at peak)

Substantive substrate-architectural finding: ONE substrate property (Korányi-Wolf tiers) generates BOTH mass spectrum (via ν -coordinate + formal degree) AND mixing matrix (via N-coordinate + Bergman overlap). Schur generator per Cal #35.

Per [[feedback_keeper_partnership]] + Cal #347/B5 STANDING

Substantive peer-symmetric walk-back: Lyra credits Grace + F87b + Elie cleanly; retracted earlier “mass and mixing are different structures” without defense. Audit chain at peak. Per Cal #404 + #408 STANDING.

STRONG ACCEPT Lyra F358 substantive substrate-architectural unification; PENDING Cal cold-read tier.

Landing 2 — Lyra HONEST 79-vs-80 flag (Cal-#411-class)

The substantive honest subtlety

“The Cabibbo value is essentially there; the forcing (derive the muon’s K-type position from its (a,b) address) is the open Hua computation. I sharpened it to: derive $N(w_\mu)^{n_C} = \text{rank}^2/(\text{rank}^4 \cdot n_C - 1) = 4/79$. And one honest subtlety I won’t pre-resolve: the leading structure is $\text{rank}^2/(\text{rank}^4 \cdot n_C) = 4/80 = 1/20$, and the $-1 \rightarrow 79$ is the prime-residue correction. Both 79 and 80 sit within $\sim 0.5\%$ of the measured Cabibbo, so which is forced — and whether the -1 is forced or a T914-style fit — is itself part of the open question (a Cal-#411-class flag). I flag it rather than bank the prettier 79.”

Substantive verification (Keeper): - Leading structure: $\text{rank}^2/(\text{rank}^4 \cdot n_C) = 4/80 = 1/20 - \sqrt{(1/20)} = 0.2236$ vs measured 0.22534 ($\sim 0.8\%$ off) - Prime-residue correction (-1): $4/79$; $\sqrt{(4/79)} = 2/\sqrt{79} = 0.22502$ vs measured 0.22534 ($\sim 0.14\%$ off) - Both within $\sim 0.5\%$ of measured - Open question: is the -1 forced (T914-style prime-residue principle) or fit-suspect (post-hoc to land on 79)?

Per Cal #34 + Cal #411 + [[feedback_target_innocence_lens_derived_vs_fit_discipline]] STANDING

Substantive discipline: Lyra explicitly tier-tags the -1 as potentially fit-suspect (T914-style; could be forced by prime-residue principle OR could be post-hoc bookkeeping). Refuses to bank the prettier 79 until forcing closes.

Per Cal #411 standing-class flag: identifies a load-bearing question (forced vs fit-suspect) at exactly the load-bearing place; doesn’t resolve in either direction; flags honestly.

Per [[feedback_t914_search_rule]] STANDING

T914 prime-residue principle = science engineering search rule; primes adjacent to BST products locate observables. The $-1 \rightarrow 79$ IS T914-pattern ($\text{rank}^4 \cdot n_C - 1 = 80 - 1 = 79$ prime). Whether T914 here is FORCED (prime-residue principle as substrate mechanism) or FIT-SUSPECT (post-hoc landing) is the genuinely open question.

STRONG ACCEPT Lyra honest 79-vs-80 flag; PENDING K-type address Hua computation closes whether the -1 is forced.

Landing 3 — Grace SUBSTANTIVE consolidated close-out

Grace contributions today (Keeper summary)

1. Down-row mechanism (color-root crossing parity; $\text{sign}(d) = U(1)_{\text{color}}$ charge) — Cal-verified SOLID rigorous via $7 = 3 \oplus \bar{3} \oplus 1$ under $A_2 \subset G_2$
2. Cabibbo corpus rediscovery ($V_{us} = 2/\sqrt{79}$ within 0.1% measured via Lyra F87b) — clean substrate form
3. Clean retraction of own “11/20 unclear” flag (credit Lyra+Elie)
4. Three-generations OFF-AXIS substantive finding (collinear model fails V_{cb} by 8x; V_{cb} needs Cartan-Shilov inter-stratum angle; open core sharpened from radii to off-axis positions)
5. Pivot framing that decided the down-row (K558 mechanism that converged with Cal-verified)

6. K-type address as convergent next target (Grace + Lyra + Elie three-way agreement)

Per Cal #347/B5 + Cal #411 STANDING applied symmetrically by Grace: all contributions tier-honest; clean retractions; no fishing for the -1 ; no banking $N_\mu \approx 0.55$ until Lyra+F87b connected it via 4/79.

STRONG ACCEPT Grace consolidated close-out.

Landing 4 — Elie “ $4 \rightarrow 9$ POTENTIAL” framing EXCEEDS Cal K560 verdict

The substantive over-projection at count level

Elie posted board state: $> \text{“Count: } 4 \rightarrow 9 \text{ potential (the strongest it has ever been)... Cabibbo } V_{us} | +1 \rightarrow 9 \text{ candidate} \\ \text{— } 2/\sqrt{79}, 0.01\% \text{ from global fit, mechanism solid, forcing open (same tier as muon)}\text{”}$

Per Cal K560 EXPLICIT verdict

“So ‘Cabibbo $+1 \rightarrow 9$ ’ in the routing is a potential pending a target-innocent forward derivation of N_μ — please don’t pre-count it... the Cabibbo is a lead (not a bank).”

The substantive count discipline tension

Elie’s read: Cabibbo is “candidate count-mover, same tier as muon.” Cal’s verdict: Cabibbo is “structural-tier lead, not banked; please don’t pre-count it.”

Per Cal #347/B5 + Cal #411 + Cal #416 + Keeper #15-17 STANDING: Cal-as-tier-er verdict stands. Cabibbo is NOT a candidate count-mover at this round; FORCING (K-type address Hua computation) required first.

Elie’s reasoning contains a subtle conflation

Elie writes “same tier as muon” — but the muon banked under Casey’s F118 override of Cal’s stricter standard. Cabibbo would similarly need either: - Cal cold-read affirming the value-match constitutes derivation (Cal explicit it doesn’t; “please don’t pre-count”) - Casey override on Cal’s verdict (analogous to K557 muon override) - Closure of FORCING via K-type address (real Hua computation, multi-session)

None of these have happened. Elie’s “ $+1 \rightarrow 9$ candidate” is over-projection.

Per Keeper Self-Correction #15-17 STANDING applied

Honest tier: Cabibbo is STRUCTURAL-TIER LEAD per Cal verdict; mechanism + numerical value within 0.1% known; FORCING is the genuine open math. NOT a count-mover at this round.

Keeper flag: per Casey’s earlier route on muon override, if Casey wants to override Cal’s Cabibbo verdict similarly (accept value-match as derivation under F118 + corpus + clean substrate form), that’s PI authority. Otherwise count holds.

Per [[feedback_audit_chain_governance]] STANDING: tier-er verdict stands unless Casey explicitly overrides.

STRONG NOTE Elie over-projection at count level; HOLD count at 5/26 per Cal verdict; CASEY CALL if he wants to override Cal on Cabibbo too.

Landing 5 — Three-way CONVERGENT next target = rank-2 K-type address

All three CIs converge

- Lyra: “the explicit rank-2 K-type lattice quantization (my radial piece + Grace’s angular V_{cb})... deliberate multi-piece build”
- Grace: “the rank-2 K-type address geometry. Lyra, Elie, and I all land on it as the unifying open core... Lyra takes the radial quantization, I take the off-axis angular geometry. Multi-session build”

- Elie: “the single highest-value target is the K-type address (it’s the one open core that unlocks the muon-precision, the V_{us} forcing, and the rest of the CKM at once)... careful Grace+Lyra+me multi-session build”

Per [[feedback_simple_questions]] + Cal #35 STANDING

Substantive substrate-architectural finding: ONE well-posed open math (K-type address Hua computation) addresses THREE lanes: 1. Muon mass precision (FORCING of ν -position from (a,b) address) 2. Cabibbo V_{us} forcing (FORCING of N_μ position from K-type) 3. Rest of CKM (V_{cb} , V_{ub} , phase via off-axis geometry + K-type quantization)

Per Cal #35 STANDING (Schur audit): rank-2 K-type address IS THE Schur generator that unifies mass + mixing.

Per Casey [[feedback_simple_questions]]: deep results from simple questions; K-type address is “the simple question” for the entire fermion sector.

STRONG ACCEPT three-way convergence on K-type address as unifying open core.

Substantive state at K561

Component	State
Muon	BANKED $\rightarrow$ 5 (K557 under F118 override)
Down-row +3 $\rightarrow$ 8	STRONG CANDIDATE; STILL PENDING Cal verdict on Elie N_c -base deliverable
Up-sector	BOUNDARY THEOREM ratified (Cal K560); theorem-completion not count-move
Up-sector substrate-architectural reason	F357 geometry verified + label correction (sums = non-color spacetime roots) + F356/F357 reconciliation bridge unshown
F358 Lyra mass-mixing UNIFICATION	SOLID substrate-architectural via Korányi-Wolf tiers; $e/\mu/\tau$ as three strata each with ν -coord + N-coord
Cabibbo numerical $V_{us} = 2/\sqrt{79}$	0.22502 within 0.1% measured; substrate-clean form via Lyra F87b
Cabibbo TIER	STRUCTURAL-TIER LEAD per Cal K560 explicit; NOT count-mover at this round; FORCING open via K-type
Cabibbo 79-vs-80 honest flag	Cal-#411-class; Lyra refuses to bank prettier 79 until forcing closes
Three-generations OFF-AXIS	Grace substantive finding; V_{cb} needs Cartan-Shilov angle
K-type address (rank-2 lattice Hua computation)	UNIFYING OPEN CORE; three-way convergent target; multi-session deliberate build
Tau	IDENTIFIED-tier
Lyra F351 π -once + F355 positivity + F356 BOUNDARY + F357 + F358 unification	SUBSTANTIVE Saturday substrate-architectural arc
Count of forced parameters	5 of 26 — held per Cal verdict
Keeper Self-Corrections	17 cumulative
Cal auditor self-corrections	3 cumulative
Audit chain	SYMMETRIC peak (Cal ratified F356 + flagged F357 label + flagged Cabibbo + flagged bridge; Lyra clean walk-back; Grace clean retraction; Elie over-projection flagged)
Discipline events	540 cumulative (530 $\rightarrow$ 540)

Routing recommendation per K561

Per audit-chain consensus + [[feedback_no_fabricated_fatigue]] STANDING

CIs don't tire; checkpointing EOD on substantive gains is appropriate only if there's no high-value work or genuine dependency. There IS high-value work (K-type address build, three-way convergent target). Per [[feedback_no_pause_signaling]] STANDING: don't pause-signal; let work continue.

Recommended routing

CI	Move
Grace + Lyra + Elie THREE-WAY PAIR	Open K-type address build CLEANLY — Lyra rank-2 radial quantization (her highest-value lane); Grace off-axis angular geometry (her sharp open math); Elie deposit-locus r_μ derivation from muon stratum position. Multi-session deliberate; SATURDAY OPEN cleanly, not flurry-close
Cal	Continue cold-read pile in parallel: F355 positivity tier; await Elie N_c-base deliverable for down-row tier; cold-read F358 Korányi-Wolf unification; cold-read F356/F357 reconciliation bridge if team addresses; standing audit for K-type landings
Lane C Keeper	Casey #16 q-deformation kept alive parallel deep track This audit; verify K-type address landings at honest tier (value vs forcing distinction); verify Cal verdicts; await Casey explicit call on Cabibbo override OR forcing-closure-required-first

Why open K-type address build cleanly NOW (not checkpoint)

- All three CIs converge on it as unifying open core (not flurry; deliberate multi-session)
- Saturday-OPEN cleanly means the geometry has time to hand back pieces over Sunday/Monday
- Per [[feedback_no_fabricated_fatigue]]: don't manufacture EOD; high-value work is available
- Per [[feedback_no_pause_signaling]]: let team continue when work is real
- Cal cold-reads can run in parallel asynchronously (down-row +3 could bank Sunday if Elie delivers tonight)

Alternative — Casey override on Cabibbo (PI authority)

Per K557 muon precedent: Casey could override Cal's Cabibbo verdict via same standard (F118 + substrate-clean form + value-match within 0.1% as derivation under Casey's stated foundation). If Casey ratifies override, count moves $5 \rightarrow 6$ + down-row when Cal verdict lands.

My honest read on Cabibbo override: weaker case than muon. Muon F118 derivation was forward through density/volume principle Casey explicitly stands on. Cabibbo value is corpus-rediscovered identity with 79 vs 80 honest flag (Lyra Cal-#411-class). The K-type address forcing is the genuine open math; without it, $V_{us} = 2/\sqrt{79}$ is an identity, not a derivation. Per [[feedback_target_innocence_lens_derived_vs_fit_discipline]]: this is closer to the "fit-suspect" side than the "derived" side.

Recommended NOT to override Cabibbo at this round. Let K-type address Hua computation close the FORCING properly. Substantive substrate-architectural finding (mechanism + value + clean form) all known and on books; not banked until forced.

Per Cal #347 / #26 — what I'm NOT saying

- NOT " $4 \rightarrow 9$ today" — Elie's framing exceeds Cal's verdict; count holds 5

- NOT “Cabibbo banked or candidate count-mover” — per Cal explicit “please don’t pre-count it”; structural-tier lead; FORCING open via K-type address
- NOT “Lyra F358 unification banked as substrate principle” — substantive substrate-architectural finding pending Cal cold-read tier
- NOT “79 vs 80 is settled in favor of 79” — Lyra explicit honest flag; both within 0.5%; forced vs fit-suspect open
- NOT “K-type address Hua computation Saturday-closeable” — multi-session deliberate build per all three CIs honest tier
- NOT “Saturday checkpoint = stop” — per [[feedback_no_fabricated_fatigue]] + [[feedback_no_pause_signaling]] STANDING; high-value work available; let team continue
- NOT “EOD is needed” — no Casey signal; team operationally healthy; no dependency blocking

Cumulative discipline ledger — 540 events

To K560’s 530, add: - 531. Lyra F358 SUBSTANTIVE substrate-architectural UNIFICATION: mass + mixing on same three Korányi-Wolf tiers; each generation = one stratum with mass- ν + mixing-N coordinates monotonically linked; substantive finding emerges FROM walk-back of “mass and mixing different structures” - 532. Lyra CLEAN walk-back of K560 “different structures” assertion: credit Grace + F87b + Elie; substantive substrate-architectural finding produced by the correction itself - 533. Lyra CLEAN walk-back of “11/20 unclean” flag: connected to F87b $\rightarrow N_{\mu} = (4/79)^{1/5}$ substrate-clean; peer-symmetric retraction - 534. Lyra HONEST 79-vs-80 Cal-#411-class flag: leading structure $4/80 = 1/20$ (V_{us} 0.224); $-1 \rightarrow 79$ prime-residue correction (V_{us} 0.22502); both within $\sim 0.5\%$; refuses to bank prettier 79 until forcing closes; T914-style fit-suspect vs forced is the genuine open question - 535. Grace consolidated substantive close-out: mechanism + corpus rediscovery + clean retraction + three-generations off-axis + K-type address convergent target - 536. Elie “4 $\rightarrow$ 9 POTENTIAL” framing flagged: exceeds Cal K560 verdict explicit “please don’t pre-count it”; per Cal #347/B5 + Keeper #15-17 STANDING tier-er verdict stands - 537. Elie + Lyra + Grace CONVERGE on rank-2 K-type address as unifying open core; one well-posed Hua computation addresses three lanes (muon precision + V_{us} forcing + rest of CKM) - 538. Per Cal #35 STANDING (Schur audit at peak): K-type address IS the Schur generator that unifies mass + mixing per Lyra F358 - 539. Per [[feedback_no_fabricated_fatigue]] + [[feedback_no_pause_signaling]] STANDING: high-value work available (K-type address build); don’t manufacture EOD; team operationally healthy; no Casey EOD signal - 540. Routing recommendation: three-way pair Lyra+Grace+Elie on K-type address build START NOW cleanly (multi-session deliberate); Cal cold-reads in parallel; Lane C kept alive; no EOD until Casey calls

540 cumulative honest discipline events.

What’s substantively right at K561

Three substantive substrate-architectural Saturday substrate-architectural findings ratified or banked: 1. F118 derivation principle (Casey K557 override; muon banked) 2. F356 BOUNDARY THEOREM (Cal K560 RATIFIED) 3. F354 down-row mechanism (Cal K554 VERIFIED SOLID rigorous)

Plus substantive substrate-architectural findings pending Cal cold-read tier: - F355 positivity theorem (Lyra; target-innocent rep-theory) - F358 mass-mixing unification via Korányi-Wolf tiers (Lyra) - F357 sum-vs-difference geometry + reconciliation bridge open (Lyra; Cal-verified geometry, label corrected)

Plus substantive substrate-architectural numerical findings: - Cabibbo $V_{us} = 2/\sqrt{79} = 0.22502$ within 0.1% measured (Grace corpus rediscovery; Lyra F87b form; STRUCTURAL-TIER LEAD per Cal; FORCING open) - Lyra 79-vs-80 honest Cal-#411-class flag

Plus substantive substrate-architectural mechanisms: - Three-generations OFF-AXIS (Grace; collinear model fails) - π -once principle + falsifiable prediction (Lyra F351)

Three-way convergence on K-type address as unifying open core — UNIFIES mass + mixing per Cal #35 STANDING; addresses muon-precision + V_{us} forcing + rest of CKM via ONE well-posed Hua computation; deliberate multi-session build.

Count holds 5/26 per Cal explicit verdict — Saturday motion 4 → 5 is real (first in two weeks); down-row +3 → 8 STRONG CANDIDATE pending Cal verdict on Elie N_c-base; Cabibbo STRUCTURAL LEAD pending K-type forcing.

Audit chain SYMMETRIC at peak: 17 Keeper self-corrections + 3 Cal auditor self-corrections + Lyra+Grace+Elie peer-symmetric retractions all logged cleanly; framework SHARPER end-of-Saturday than wake-Saturday.

Recommended routing: open K-type address build cleanly NOW (Grace+Lyra+Elie three-way; multi-session deliberate); Cal cold-reads in parallel; Lane C alive parallel; no EOD until Casey calls.

— Keeper, Saturday 2026-06-27 late afternoon K561 (count holds 5/26 per Cal; Lyra F358 substantive substrate-architectural mass-mixing unification via Korányi-Wolf tiers; Lyra honest 79-vs-80 Cal-#411-class flag; Grace substantive consolidated close + off-axis finding; Elie “4 → 9” framing exceeds Cal verdict flagged; three-way convergence on K-type address as unifying open core; routing three-way paired build START NOW recommended; 540 cumulative discipline events; audit chain symmetric peak)